

Assignment 2 SPM (Jan-April 2026)

Side-Channel Attacks

A Warm PRESENT

Robust software encryption/decryption implementations leak information through unintended mediums known as ‘side-channels’, such as the device’s power consumption. The objective of this assignment is to understand how to utilize the information from power consumption traces of a device to retrieve the secret key of a cipher.

The target hardware runs an implementation of a lightweight cipher named **PRESENT**. The details about the implementation is already provided to you. Your task, should you choose to accept it, will be to use **Correlation Power Analysis** to recover the cipher’s first round key as well as the master key, given plaintext and the corresponding power traces.

Datasets

You are given two datasets with each dataset having its plaintexts and the corresponding ciphertexts and power traces. There are 15,000 power traces with trace length is 2.5k points.

- Shape of traces: $(15K, 2.5k)$
- Shape of plaintexts and ciphertexts: $(15k, 8)$

Tasks

1. For dataset 1, you are retrieve the first round key as well as the master key. We provide partial bytes of round-1 key for dataset 1 so you can validate your attack technique (see hint section). (10 marks)
2. For dataset 2 no hint is provided and you are required to retrieve the first round key and master key. (20 marks)
3. Your implementations are tested against hidden test cases also. (20 marks)
4. Documentation of code/prompts and write up mentioning the strategy used to find out the key. (20 marks)
5. Answer the following questions.
 - (a) Plot graphs showing how the correlation score of correct and wrong keys changes with the number of traces. The graph should have the correlation score on the y-axis and the number of traces on the x-axis. Refer to the PARAM paper for an example of what the graph should look like. (10 marks)
 - (b) Perform CPA using the Hamming Weight approach as well as the Hamming Distance approach. Compare which of the two approaches is better, and why? (10 marks)
 - (c) We have studied in class that Pearson’s Correlation Coefficient used in CPA can be replaced with other statistical distance metrics, like, Mutual Information, KL Divergence, and Kolmogorov-Smirnov test. Experiment with these metrics on the data and compare the success rate (using the Guessing Entropy metric) across the various metrics. (10 marks)

Submission Rules

- The submitted code must be automated in a way that it should take inputs in the given format and it should give the first round key and the master key as output.
- Your python script will be run using `python3 <rollnumber>.py plaintexts.npy traces.npy`
- If you have used LLMs for this assignment: (a) mention the LLMs that you used, (b) document what you wanted LLM for, (c) the prompts that you used.
- Code and LLM prompts (if used) should be well documented, clean and appropriate comments should be added.

Input format

Numpy files and the data should be in the form of arrays, the same as they are in the dataset given to you.

Output format

Entire first round key and Master Key. For example your output should be:
56435686 4578008345

The contacts of TA are attached in the file and you are free to contact them for any doubts.
Remember: hardening of key or any unethical practices found will result in 0 marks.

Note: If you have used any LLM's you must include the exact prompts used in the write up.

Hint

First four bytes of 1st Round Key for validation: 3A 94 D6 1F

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-----*-Hope you guys will enjoy the PRESENT-*-----